# SARA BOURAS

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## STATE ENGINEERING STUDENT IN ARTIFICIAL INTELLIGENCE & BIG DATA

Final-year engineering student at HESTIM, specializing in Artificial Intelligence and Big Data. Strong skills in Data Science, Machine Learning and Business Intelligence, with hands-on end-to-end experience: data collection and cleaning, predictive modeling, deployment and visualization. Seeking a graduation internship (PFE) in Data Science / Machine Learning.

## EDUCATION

**HESTIM** Oct. 2022 – Present
*State Engineering Degree in Artificial Intelligence and Big Data — 3rd year*

**LAMARTINE** June 2021
*Baccalauréat in Mathematical Sciences — Option A*

## PROFESSIONAL EXPERIENCE

**Groupe Construlting Sigman — Casablanca, Morocco** May – July 2026 (3 months)
*Business Intelligence / Data Intern*

- Designed and deployed two decision-support systems for recruitment and treasury, from data entry via Power Apps to SharePoint storage and Power BI visualization.
- Cleaned and transformed heterogeneous Excel data with Power Query/M and built KPIs with DAX; automated data collection and refresh via Power BI Service.
- Developed a logistic regression model in Python to estimate application success from external data, handling class imbalance and feature selection.

**Royal Air Maroc — Casablanca, Morocco** June – July 2025 (1.5 months)
*Web Development & Testing Intern*

- Built an automated testing web application with Python, Selenium WebDriver, SQLite, HTML/CSS and JavaScript, allowing users to select or write scripts and choose the browser for execution.
- Automatically ran test scenarios across different pages of the website, with a detailed test-results reporting system.

## PROJECTS

**Autonomous Inspection Robot — Post-Earthquake Damage Assessment** Oct. 2026 – Present

- Robot inspecting an earthquake-damaged building: Gazebo world and SLAM Toolbox mapping; Nav2 and YOLOv8 in progress.

**AI Business Briefing — Automated Business Insights Generation** In progress
*Technologies: React, TypeScript*

- Building a tool that turns Excel/CSV data into an automated summary: KPI trends, significant variations, potential causes and points of attention.

**Medical Multi-Agent AI System** March – April 2026
*GitHub: SiriusRalph/Projet-agentic-AI | README: full documentation | Technologies: Python, LangGraph, LangChain, FastAPI, React, Ollama, MCP*

- Designed a multi-agent AI system (4 specialized agents) simulating preliminary clinical triage, orchestrated via LangGraph with clinical tool integration through MCP.

**Etudafrik — Intelligent School Guidance Platform** May 2026
*GitHub: SiriusRalph/etudafrik | Technologies: React, TypeScript, Supabase, Grok API*

- Built a web platform centralizing information on private Moroccan schools, allowing African students to search and apply from a single interface.
- Designed a recommendation system via the Grok API that matches schools to the student's profile; built dedicated dashboards for students, schools and administrators.

**IoT Hospital Intrusion Detection System** April 2026
*GitHub: SiriusRalph/iomt-intrusion-detection | README: full documentation | Technologies: Python, Pandas, Scikit-learn, Streamlit, Raspberry Pi*

- Detected IoT intrusions on 200,000 CICIoT2023 samples, with preprocessing, feature engineering and comparison of 7 classification models; optimized decision tree: 88.17% accuracy, 96% recall.
- Deployed the model and a real-time monitoring Streamlit dashboard on Raspberry Pi, optimizing data transmission to reduce energy consumption.

## SKILLS

**Programming:** Python, SQL
**Data & ML:** Pandas, NumPy, Matplotlib, Machine Learning
**Web:** React.js, HTML, CSS, JavaScript
**Databases:** MySQL, MongoDB, Supabase
**BI & Tools:** Power BI, Power Query, Power Apps, SharePoint, Excel, GitHub
**Languages:** French (fluent), English (fluent), Arabic (native)

## CERTIFICATIONS

- **IBM Data Science Professional Certificate** (Coursera / IBM) — Python and SQL for data science, data cleaning and visualization, data science methodology and building Machine Learning models.
- **Machine Learning** (Coursera) — Supervised and unsupervised learning algorithms, model evaluation and applied Machine Learning best practices.
- **Oracle Database Foundations** (Coursera / LearnQuest) — Relational databases, table design and SQL queries in Oracle.
- **Project Management** (Coursera) — Project planning, risk management and agile methodologies.